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Comprehensive time-series analysis of bridge deformation using differential satellite radar interferometry based on Sentinel-1

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ABSTRACT

We present a comprehensive methodological framework for structural deformation monitoring of critical infrastructure assets based on differential SAR interferometry. By employing persistent scatterer interferometry, deformation time series in line-of-sight are derived from freely available Sentinel-1 single look complex products. These raw time series are analysed and refined using an extensive post-processing chain to obtain daily rates for vertical and horizontal deformation components. The post-processing includes cleaning, smoothing and a temperature correction to account for different sensing times in ascending and descending orbits. Longitudinal clustering of time series is used to reveal spatial patterns in the single epoch deformation series. Seasonal trend decomposition of the aggregated time series is performed to separate deformation trends from seasonal deformations. The applicability of the framework is showcased at the example of an integral concrete bridge located in the port of Vienna. Results are validated against in situ deformation measurements. The presented framework constitutes a blueprint for the continuous monitoring of critical infrastructure assets using satellite interferometry, which may supplement conventional structural health monitoring.

1. Introduction

Modern society heavily relies on a multitude of critical infrastructure, which is essential for the functioning of society and economy (UNISDR, 2009; Council of European Union, 2008). In particular, engineering structures protrude as constructions of specific importance, since these physical assets play a vital role in a wide variety of sectors, including for e.g. transportation systems, public protection, energy, water supply or communication systems (NIPP, 2013; PPD-21, 2013; Council of European Union, 2008). Any disruption of such critical systems may cause serious impacts on society. Their incapacitation or destruction might entail debilitating effects on security, safety or health, or any combination thereof (Schlögl et al., 2019).

Consequently, monitoring the structural safety of critical infrastructure is of major importance. In the context of structural health monitoring, the continuous measurement of structural deformation does play an essential role (Bao et al., 2019; Luo et al., 2019).

Recently, methodological advances in differential interferometric synthetic aperture radar (DInSAR) analysis (Crosetto et al., 2016) and an increased availability of satellite radar data (Devanthéry et al., 2016) have sparked interest in using large stacks of interferometric SAR images to support critical infrastructure monitoring. With the launches of Sentinel-1A (April 2014) and Sentinel-1B (April 2016), C-band SAR data (centre frequency: 5.405 GHz) became freely available across large parts of the globe. Sentinel-1 single look complex (SLC) products are characterised by a high spatial resolution of 5 × 20 meters and a high temporal resolution. The constellation of the two satellites entails a repetition time of 6 days, and resulting data are provided near-real time. The temporal resolution can even be increased by combining data from overlapping orbits (ESA, 2020a; ESA, 2020b).

Concerning aforementioned methodological advances, persistent scatterer interferometry (PSI; Ferretti et al., 2001) represents an important specific class of DInSAR analysis. PSI is a powerful remote sensing method for measuring and hence monitoring surface displacements on the Earths surface over time. By exploiting information contained in the measured phase values of co-registered, multi-temporal SAR imagery, ground deformation and corresponding displacement rates can be obtained with sub-centimetric precision. Thus, the core aim of PSI is the derivation of deformation time series estimated over the analysed permanent scatterers by separating the displacement phase component from multiple SAR images acquired over the same area.

PSI analyses have been used in a wide variety of surface deformation

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monitoring efforts. Crosetto et al. (2016) summarise SAR applications into four main classes, namely (i) urban, peri-urban and built; (ii) subsidence and uplift; (iii) landslides; and (iv) geophysics. Albeit there are various studies belonging to the latter three classes, applications related to infrastructure deformation monitoring are comparatively rare and rather new. In particular, studies using Sentinel-1 data are still scarce, but are gaining ground recently, especially for monitoring movements in urban contexts (Morishita et al., 2020; Scoular et al., 2020; Venmans et al., 2020).

While PSI is generally applicable on a variety of different structures, we focus on transportation infrastructure monitoring. Due to the high importance of mobility in everyday life, transportation infrastructure assets (and their users) are particularly vulnerable to structural failure. This is highlighted at the tragic example of Morandi bridge across the Polcevera river in Genoa, whose collapse caused the death of 43 people (Milillo et al., 2019).

Bridges are particularly prominent assets of land transportation infrastructure. Apart from the obvious catastrophic effects of structural failure, unplanned closures due to repair, upgrading or maintenance work may also impose substantial impairment of traffic fluidity, thereby leading to tangible and intangible losses. Therefore, monitoring of (increasingly ageing) bridge infrastructure requires attention from engineers and managers alike to safeguard safe, efficient and effective infrastructure operation.

While traditional methods of on-structure bridge surveying provide only limited and point like deformation information, which is often time consuming and manpower demanding, InSAR may facilitate coverage of wide areas with a high frequency, saving time and money and potentially mitigating hazards (Huang et al., 2017; Qin et al., 2017; Qin et al., 2019; Milillo et al., 2019).

The application of multi-temporal SAR interferometry for bridge monitoring is a relatively new area of research, with only a handful of dedicated publications over the last decade. A vast majority of existing work is based on high resolution X-band SAR data, such as TerraSAR-X and COSMO-SkyMed.

Several of these studies are motivated by specific cases of bridge failure. One of the first studies investigating a bridge collapse using SAR data was conducted by Sousa and Bastos (2013), who applied a combination of PSI and Small Baselines methods using ERS data. They were able to detect substantial deformations of about 20 mm yr^{-1} in line-of-sight direction for an investigation period of 5.5 years prior to the collapse. One year later, Bakoň et al. (2014) showed that deformations of a bridge following the crash of a tug could be registered with PSI using Envisat data. Selvakumaran et al. (2018) showed that in the case of a partial collapse of a masonry arch bridge due to scour, unusual deformations could be identified for one point one month prior to the collapse through a small baseline subset (SBAS) approach employed on TerraSAR-X data using outlier detection. As mentioned before, Milillo et al. (2019) analysed the collapse of the concrete cable-stayed Morandi bridge in northern Italy, using COSMO-SkyMed, Sentinel-1 and Envisat data. Multi-temporal InSAR was applied and a Markov chain Monte Carlo approach was used to solve for the three-dimensional deformation of the bridge deck. Envisat data exhibited insufficient spatial resolution and repeat time for surveying the deck level, thereby also not revealing any long-term trends for the bridge pillars. Sentinel-1 data merged with COSMO-SkyMed data proved to be suitable for three-dimensional deformation decomposition.

In recent years, feasibility and applicability of DInSAR for structural health monitoring have been explored, with a particular focus on thermal expansion. Earlier works include those by Gernhardt et al. (2010) on the potential of high-resolution SAR data for monitoring man-made infrastructure in urban areas, by Monserrat et al. (2011) on analysing the thermal dilation phase component within persistent scatterer interferometry, by Reale et al. (2013) by distinguishing linear deformation and thermal dilation of buildings, and by Goel et al. (2014) on the investigation of different DInSAR techniques for thermal dilation monitoring of complex urban infrastructure. Early bridge-specific applications are presented by Fornaro et al. (2013), who assessed the thermal response of the Musmeci bridge in Potenza, Italy and by Lazecký et al. (2015), who demonstrated the potential for monitoring deformations and deformations due to thermal dilation of bridge structures using PSI at the example of three case studies in Bratislava (Slovakia), Ostrava (Czech Republic) and Hong Kong. The assessment of thermal expansion of bridges is further investigated by Lazecky et al. (2017), who presented a method to refine the thermal expansion component in order to improve point coherence and support the determination of the linear deformation of bridges.

Huang et al. (2017) used PSI based on Sentinel-1A data for monitoring a long-span railway bridge across the Yangtze (China). They were able to discern displacement patterns due to thermal expansion of the bridge based on a strong correlation between longitudinal displacement and ambient temperature. Cusson et al. (2018) investigated PSI displacement rates of a highway bridge in Ontario, Canada. They found that the railings of the bridge under consideration turned out to be excellent natural reflectors, resulting in a large number of scatterer points. Resulting deformation rates were compared with measurements from temperature sensors in order to remove the thermal contribution and allow focusing on mechanical deformations. Results were further validated against measurements from displacement sensors and numerical modeling thermal data.

Qin et al. (2017) used TerraSAR-X data for multi-temporal SAR interferometry of bridges and analysed the correlation between temperature and resulting displacement as well as structural mechanics. This work was advanced in Qin et al. (2018) by focusing on the discrimination between surface deformation and thermal dilation of arch bridges, and in Qin et al. (2019), where a bridge-tailored multi-temporal DInSAR approach monitoring complexly structured bridges is presented. In this last study, the authors present a coherence-driven point-like target selection strategy as well as multi-track result 3D visualisation, damage sensitive point identification and target point classification.

As opposed to application oriented studies, a comparatively small number of publications (explicitly) focuses more on methodological innovations demonstrated – among other examples – at bridges.

Kang et al. (2017) proposed a general framework for object-based InSAR parameter retrieval, where the parameters of the whole object are jointly estimated by the inversion of a regularised tensor model instead of pixelwise. They apply the robust object-based multipass InSAR deformation reconstruction to bridge monitoring in the central area of Berlin. It is shown that by integrating semantic and geometric information into geophysical parameter retrieval the accuracy of estimates can be improved and the number of images required can be reduced. In addition, they noted that the bridge area undergoes a periodic seasonal motion that is primarily caused by the thermal dilation of the steel railways on the bridge.

Duan et al. (2018) used a simulated annealing algorithm to achieve high precision deformation results. Apart from a simulation experiment, the methodology was also tested using TerraSAR-X images of Beijing (China), with a special focus on linear deformation results of the Wenyuhe Bridge. An application of the simulated annealing algorithm to bridge measurements showed good agreement to trends measured by leveling data.

Finally, InSAR analyses for infrastructure monitoring have been used in a broader context.

Hoppe et al. (2016) investigated the feasibility of InSAR analysis based on civilian radar satellites for longterm performance monitoring of transport infrastructure. Using data from COSMOSkyMed, they demonstrated specific applications, including sinkhole detection in karst terrain, slope stability monitoring and infrastructure assessment, concluding that satellite radar remote sensing provides good alternate or complementary diagnostic tools for monitoring the transportation network. Ozden et al. (2016) evaluated the use of DInSAR for pavement

monitoring and concluded that SAR-based methods are useful as a complementary tool for pavement surface deformation detection rather than a replacement for current technologies and practices.

Zhao et al. (2017) used a long-short baseline iteration method to tackle phase-wrapping problems in complex areas stemming from large elevation differences between the top and bottom parts of large structures – especially when combined with a long interferometric baseline. Specifically, groups of interferometric pairs with short baselines were processed to obtain rough initial elevation estimations of the persistent scatterers at Lupu bridge, a 100 m tall arch bridge in Shanghai. The method resulted in improved estimated deformation rate accuracy. Both linear and seasonal deformations could be extracted and symmetric progressive deformations were found, agreeing with the bridge design.

Delgado Blasco et al. (2019) used PSI based on Sentinel-1 data to monitor urban subsidence in the Rome metropolitan area (Italy), including an assessment of ground displacements on urban infrastructures such as road networks, buildings and bridges.

Fárová et al. (2019) used Sentinel-1 to monitor highway stability in the Central Bohemian Highlands, where a massive landslide had occurred during highway construction in 2013. They found that both single-pair DInSAR and PSI were suitable for detecting vertical displacement hotspot sections. A validation of results from the SAR interferometry against in situ measurements (geodetic measurements, 3D inclinometers, laser scanning) showed that deformation rates derived via PSI corresponded well with the in situ measurements.

Even though the accuracies of modern X-band SAR sensors are currently unrivalled – mainly due to their high spatial resolutions – Sousa et al. (2016) emphasise the potential utility of C-Band SAR interferometry for critical infrastructure monitoring at the example of dam monitoring.

This study explores the capabilities of using freely available Sentinel-1 radar data for supporting on-site structural measurements of transportation infrastructure assets. An extensive post-processing framework for the analysis of critical transportation infrastructure is presented and demonstrated at the example of an integral concrete bridge located in the port of Vienna. Results are validated against in situ deformation measurements taken over an extensive time period.

2. Data and methods

2.1. Study area

While the PSI analysis itself was performed for a larger area covering the city of Vienna and its surroundings (see supplement 1), PSI time-series post-processing was only conducted for a small subset of PS points covering a selected infrastructure asset.

The study object *Seitenhafenbrücke* ('side harbor bridge') is an integral concrete bridge (i.e. without bearings and expansion joints in its superstructure) across the Danube Canal (Fig. 1A), a former arm of the river Danube. It is located in the port of Vienna at N48°10′37.2″/E16°2750.4″ (Fig. 1B) and has been opened in 2011. While it is mainly used for individual traffic and the operation of the port of Vienna, it also features a path for pedestrians and bicycles. The bridge is a slender construction with a length of 128.7 metres and a width of 15 m. It consists of two river bank structures, each with a span of 32 m and a river structure with a span of 64.7 m. Pairs of steel supports are arranged on both banks of the Danube Canal, sitting on specially shaped cast steel nodes (Fig. 2). For details on structural engineering see (Kral et al., 2012 and Geier et al., 2014).

2.2. Data

2.2.1. Sentinel-1 SLC

The PSI analysis is based on Sentinel-1 SLC data products. The time period under investigation covers three full years and ranges from October 2016 to October 2019. Data is available from four different orbits (Table 1).

2.2.2. GACOS

Atmospheric correction of InSAR measurements was performed using high spatial resolution tropospheric delay correction maps for SAR interferograms available through the Generic Atmospheric Correction Online Service for InSAR (GACOS). These zenith total delay maps are based on the iterative tropospheric decomposition model by Yu et al. (2017), which integrates high spatial (0.125 degrees) and temporal (6 h) resolution ECMWF weather model data as well as SRTM DEM and ASTER GDEM data to separate stratified and turbulent signals from tropospheric total delays (Yu et al., 2018a; Yu et al., 2018b). GACOS data sets used for correcting interferograms for atmospheric effects are derived for every Sentinel-1 acquisition according to their location and sensing time.

2.2.3. In-situ data

The bridge is monitored using a comprehensive, tailor-made in situ monitoring system, which draws on proven components (Geier et al., 2014). The system measures several characteristic variables, namely the vertical deformation of the bridge (measurement device: electronic water level gauge), the earth pressure at the abutment wall (earth pressure sensors), the length change of the bridge (distance laser sensors), the concrete temperature (PT100 temperature sensors) as well as the angular inclination at the steel columns and the abutment wall (inclination sensors). All measurements are available for the time period between December 2011 and June 2017. Location of the deformation sensors is shown in Fig. 2.

2.3. Methods

The whole processing chain described in this study consists of three



Fig. 1. (A) The study object *Seitenhafenbrücke* (Wikimedia Commons, 2011) and (B) its location in Vienna.

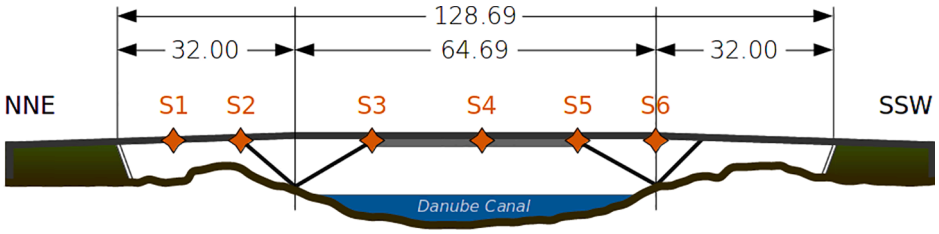


Fig. 2. Longitudinal section of *Seitenhafenbrücke* and position of in situ sensors. S1 to S6 denote the six water level gauge measurement points.

Table 1
Overview of Sentinel-1 SLC data used in this study. The number of PS-points refers to the number of points within the area of interest as depicted in Fig. 4.

orbit	orbit direction	sensing time	start date	end date	images	leader date	PS-points
22	descending	05:09 UTC	2016–10-02	2019–10-05	179	2018–03-14	75
73	ascending	16:42 UTC	2016–10-05	2019–10-02	180	2018–02-09	53
124	descending	05:02 UTC	2016–10-03	2019–10-06	180	2018–03-15	88
146	ascending	16:51 UTC	2016–10-04	2019–10-01	180	2018–03-04	84

independent workflows, which are all carried out using open source software. First, single leader InSAR processing is performed using ESA's *Sentinel Application Platform* (ESA SNAP) and the Python wrappers available through *snap2stamps* Delgado Blasco et al. (2019), respectively. Second, PSI processing is based on the *Stanford Method for Persistent Scatterers*, using the StaMPS implementation written in MATLAB (Hooper et al., 2007). Third, the extensive post-processing is implemented in R (R Core Team, 2020).

Since descriptions of the detailed methodological workflows of SNAP and StaMPS PSI (pre) processing are abundant in literature (e.g. Delgado Blasco et al., 2019; Fárová et al., 2019; Hooper et al., 2018; Hooper et al., 2007), we only provide a general overview of the first two pipelines and focus specifically on the innovative aspects of the extensive post-processing workflow.

2.3.1. PSI processing

At first, a suitable single leader scene, with which the interferograms are to be formed, was selected. This leader scene was chosen by maximizing the sum correlation, which is characterised by time interval, perpendicular baseline, difference in Doppler centroid and thermal noise (Zebker and Villasenor, 1992). The leader and supporter scenes were then split in order to reduce time and processing efforts, and orbital corrections were applied. Subsequently, coregistration was performed using back-geocoding and the one arcsecond SRTM DEM. Eventually, interferograms were computed, and flat-earth phase (i.e. the phase contribution due to Earth's curvature) and topographic phase contributions were removed.

The actual PSI analysis commenced after these single leader InSAR preprocessing steps, according to the processing chain described in Hooper et al. (2018). By utilizing the amplitude dispersion index, which indicates an estimation of the phase stability, a preselection of potential PS pixels was performed. This subset of pixels, including the PS candidates, was then tested for phase stability using phase analysis in order to estimate the probability of each pixel being a PS pixel (Hooper et al., 2007).

StaMPS PS selection is based on phase characteristics of potential PS pixels. Pixels were selected according to the maximum acceptable percentage of random phase. Most random-phase pixels were dropped and pixels adjacent to PS pixels resulting from the same dominant scatterer were weeded based on their phase noise level. Phase correction was performed by correcting the wrapped phase for spatially-uncorrelated look angle errors. Subsequently, the wrapped phase – which is given in modulo 2π – was unwrapped in order to retrieve the distinct phase. Unwrapping was performed in three dimensions: two in space and one in time (Hooper et al., 2007). Eventually, spatially correlated look angle errors, which are mostly due to spatially-correlated DEM errors, were estimated and subtracted together with leader atmosphere and orbit errors (Hooper et al., 2018).

In order to mitigate the topography-correlated atmospheric phase contribution, atmospheric correction was performed by employing the *Toolbox for Reducing Atmospheric InSAR Noise* (TRAIN) and using data from the *Generic Atmospheric Correction Online Service for InSAR* (GACOS; Yu et al., 2018a; Yu et al., 2018b; Yu et al., 2017).

This procedure was carried out for all four orbits under consideration, using the mean value for the whole area as reference value.

2.3.2. Post-processing

In order to eventually obtain robust estimates of bridge deformation that could be compared to in situ measurements, the resulting PS time series obtained from PS-processing through StaMPS/MTI and subsequent reduction of atmospheric InSAR noise via TRAIN were processed further.

The general workflow building upon PS time series can be summarised as follows:

1. **Time series cleaning**: In order to remove outliers, time series obtained after running GACOS through TRAIN were cleaned. This was achieved by using Friedman's Super Smoother (Friedman, 1984) for non-seasonal series and a robust STL (Seasonal decomposition of time series by LOESS, c.f. Cleveland et al. (1990)) decomposition for seasonal series. Only extreme values outside $\pm 3 \times (q_{0.75} - q_{0.25})$ (i.e. three times the interquartile range) are labelled as outliers. To estimate missing values and outlier replacements, linear interpolation was used on the (possibly seasonally adjusted) series (Hyndman and Khandakar, 2008).
2. **Time series smoothing**: After outlier removal, resulting time series are smoothed to eliminate random fluctuations and obtain a clearer signal. Four different smoothing methods were compared, namely locally weighted scatterplot smoothing (LOESS; Cleveland et al., 2017), generalised additive models (GAM; Wood et al., 2016; Hastie and Tibshirani, 1986), Friedman's Super Smoother (Friedman, 1984) and Whittaker smoother (Eilers, 2003).
3. **Time series clustering I**: In order to group PS points showing similar behaviour over time, smoothed time series were clustered using *k*-means for longitudinal data (i.e. trajectories, Genolini et al., 2015). This was done by using all smoothed PS time series for all four orbits conjointly. In order to determine the optimal number of clusters, different cluster sizes for all integers between 2 and 6 clusters were tested 20 times each. This was done to check consistency of the results across the time series obtained from the four different orbits.

4. **Bias correction**: The different sensing times of descending (in the morning around 05:00 UTC) and ascending (in the evening around 17:00 UTC) orbits result in a slight systematic error. Differences between evening and morning measurements taken on the same day show a distinct pattern. This is illustrated by in situ deformation measurements taken every six hours between January 2012 and July 2017 at six different sensor locations along the bridge using a water level gauge. Specifically, it can be observed that there is no constant shift along the vertical axis, but rather a seasonal one, with differences being larger during summer than in winter (Fig. 3). As opposed to a constant shift throughout the year, which could be ignored due to DInSAR measurements being relative to the leader date rather than absolute measurements, a seasonally dependent shift does introduce a systematic bias when combining multiple orbits to derive horizontal and vertical components from line-of-sight data.

   This bias is accounted for by applying a correction for each month, which is dependent on the prevailing environmental conditions. The correction relies on aforementioned in situ deformation measurements of vertical bridge deformation. Based on these in situ measurements, observed deformation differences between 03:00 UTC and 15:00 UTC (which are the two in situ measurements closest to the acquisition time of Sentinel-1 SLC products) were computed for each day. These deformation differences were then combined with hourly weather data on air temperature, precipitation, freezing level (i.e. the zero degree isotherm), relative humidity, wind direction, wind speed, global solar radiation and sunshine duration at both 03:00 and 15:00. Random forest models (Wright and Ziegler, 2017) explaining deformation difference through these 16 meteorological variables were trained for each of the six sensors. Hyperparameter tuning was done for each of the six models separately by performing an extensive grid search, which was evaluated by means of fivefold cross validation using the root mean squared error as performance measure.

   The random forest model trained with optimal hyperparameters was subsequently used to predict daily deformation differences for the time period under consideration for PSI analysis (i.e. October 2016 to October 2019), resulting in time series of estimated daily vertical displacement *differences* at the location of each of the six sensors. These time series of vertical displacement differences were then transformed into displacement differences in line-of-sight for descending orbits 22 and 124. This is achieved via $d_l = \frac{d_v}{\cos(\alpha)}$, with d_l denoting displacement in line-of-sight, d_v denoting modelled vertical displacement and α denoting the incidence angle. Resulting time series of daily correction values for the two descending orbits were then smoothed using LOESS with $\alpha = 0.25$, corresponding to the span also used for smoothing the deformation time series in step 2. Eventually, time series of PSI measurements in both descending orbits were corrected by adding the time series of smoothed bias correction values in line-of-sight for the closest sensor location.
5. **Line-of-sight to vertical**: For each orbit, deformation time series are available as one-dimensional projection onto the satellite's line-of-sight. However, motion usually occurs in all three dimensions (Aslan et al., 2020). By combining ascending and descending PSI measurements, a conversion from line-of sight to vertical and horizontal components can be computed according to Eq. (1) (c.f. Delgado Blasco et al., 2019):

$$\begin{bmatrix} d_v \\ d_{h_{ald}^{dsc}} \end{bmatrix} = \begin{bmatrix} \cos\theta^{asc} & \frac{\sin\theta^{asc}}{\cos\Delta\alpha} \\ \cos\theta^{dsc} & \sin\theta^{dsc} \end{bmatrix}^{-1} \times \begin{bmatrix} d_{LOS}^{asc} \\ d_{LOS}^{dsc} \end{bmatrix} \tag{1}$$

   where d_{LOS} is the displacement along the line-of-sight, d_v is the vertical displacement component (i.e. up/down) and $d_{h_{ald}^{dsc}}$ is the projection of the horizontal displacement in descending azimuth look direction (ALD), which corresponds approximately to deformations obtained in East/West direction, with a slight clockwise shift of 14.6 degrees. Coincidentally, the study object is oriented from ENE to WSW, which results in the descending ALD being nearly orthogonal to the bridge orientation. θ denotes the incidence angle and $\Delta\alpha$ is the satellite heading difference between ascending and descending orbit.

   An artificial grid of 0.00001 × 0.00001 degrees (this corresponds

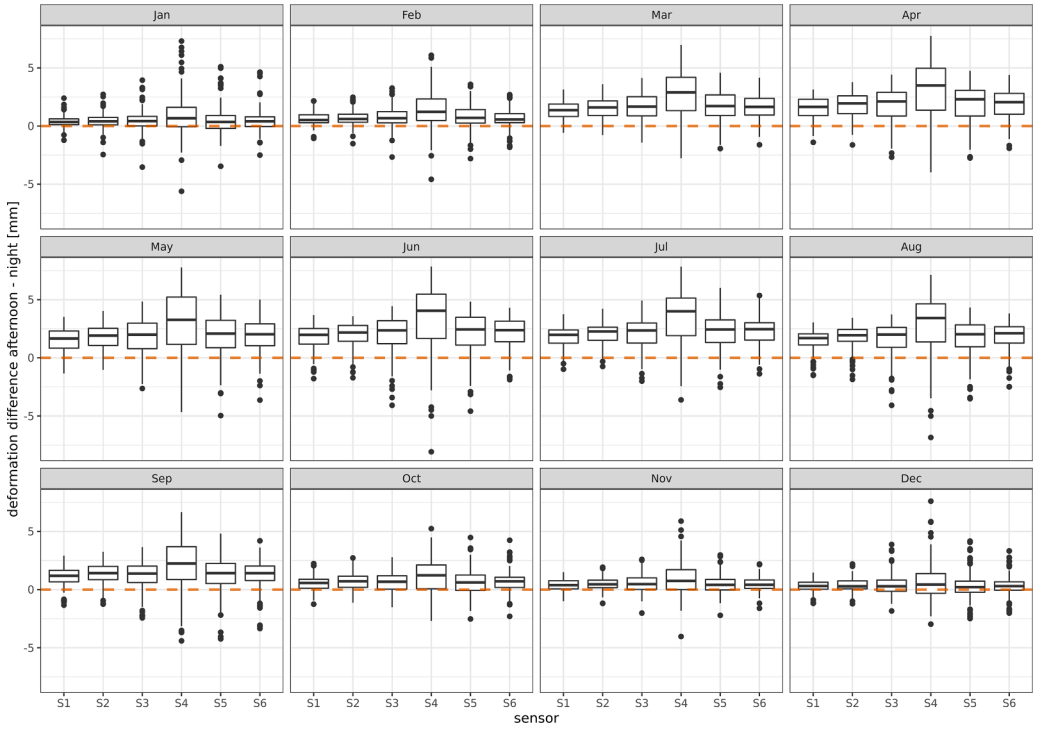


Fig. 3. Boxplots displaying the difference of measured bridge deformation between afternoon (15:00) and night (03:00). Measurements reveal a seasonal pattern, with differences being more pronounced during summer.

to a grid size of approx. 0.75 m × 0.75 m) was created across the bridge. For all grid points, the nearest ascending and descending scatterer points within a radius of 10 meters were used to perform the conversion. All other grid points were dropped.

6. **Time series clustering II**: Both vertical and horizontal deformation component series were clustered using *k*-means for longitudinal data in order to identify consistent deformation patterns within the final combined deformation values. The approach is identical to (3) described above, only the input data set is different.
7. **Aggregation of results**: The derivation of final, aggregated deformation patterns was based on two main considerations. Based on the workflow so far there were two obvious approaches to derive ensembles of deformation patterns. First, deformation time series could be aggregated per time series cluster. Second, deformation patterns could be aggregated using sets of PS-points within a given radius around the six in situ sensors. While the first approach allows for the assessment of PS-points behaving consistently over time (which *may* also coincide with respect to spatial patterns), the second approach provides a longitudinal section across the bridge, and facilitates comparison with in situ measurements.

   Eventually, it turned out that by combining both approaches, the respective benefits of each one can be harnessed. Aggregated deformation patterns were therefore derived as ensembles of deformation time series with respect to (i) the closest sensor within a given radius and (ii) corresponding time series clusters. Ensembles of the most representative suitable cluster per sensor were then compared to in situ measurements for validation purposes.
8. **Time series decomposition**: The final time series are decomposed into seasonal component, a trend component and random noise by applying STL ('Seasonal and Trend decomposition using Loess') decomposition (Cleveland et al., 1990).

3. Results

In order to maintain consistency with the methods section and to provide a concise description of results for each of the post-processing steps, results are structured according to the post-processing workflow outlined in Section 2.3.2.

1. **Time series cleaning**: The outlier identification and replacement procedure produced consistent and robust results for all four orbits. Clear outliers in both seasonal and non-seasonal time series were removed, whereas smaller deviations were not eliminated, thereby still adequately representing the uncertainty (noise) stemming from StaMPS PSI processing.

   A complete overview of the time series cleaning results for all deformation time series are available in supplement 3.
2. **Time series smoothing**: Local polynomial regression fitting using second degree polynomials with a span of $\alpha = 0.25$ was selected to smooth the scatterplots. This LOESS method was chosen due to its excellent fit while still exhibiting acceptable computational performance.

   A complete overview of the time series smoothing results for all cleaned deformation time series are available in supplement 4.
3. **Time series clustering I**: Despite the bias introduced by different sensing times and deformation values in line-of-sight which vary due to different incidence angles, time series clustering worked quite well already on the cleaned and smoothed deformation time series (Fig. 4). Four distinct clusters could be identified: The green cluster (A) showed no distinct pattern and represented noisy PS points. The orange cluster (B) was somewhat similar to (A), but with a more pronounced seasonal pattern of deformations towards the satellite (+LOS) in winter and away from the satellite in summer (-LOS). The violet cluster (C) corresponded to PS points directly on the bridge and showed a very consistent picture. It was characterised by a clear seasonal deformation, with deformation towards the satellite in late summer and deformations away from the satellite in winter. Finally, the pink cluster (D) did only comprise two points at the very northeast of the stretch of road under consideration, with a strong, consistent downward deformation.
4. **Bias correction**: The random forest models showed solid performance in modeling deformation differences between different sensing times based on weather variables, exhibiting an average out-of-bag (OOB) RMSE (across all six sensors) of 0.87 mm. Errors varied slightly between sensors, with larger OOB RMSE for models in the middle of the bridge and the southern part of the bridge (S1: 0.53 mm, S2: 0.58 mm, S3: 0.88 mm, S4: 1.44 mm, S5: 1.01 mm and S6: 0.79 mm).

   A comparison of predicted versus observed deformation differences between 15:00 UTC and 03:00 UTC (Fig. 5) indicated a slight bias towards underestimating particularly high deformation differences, while the majority of observations was modelled well.

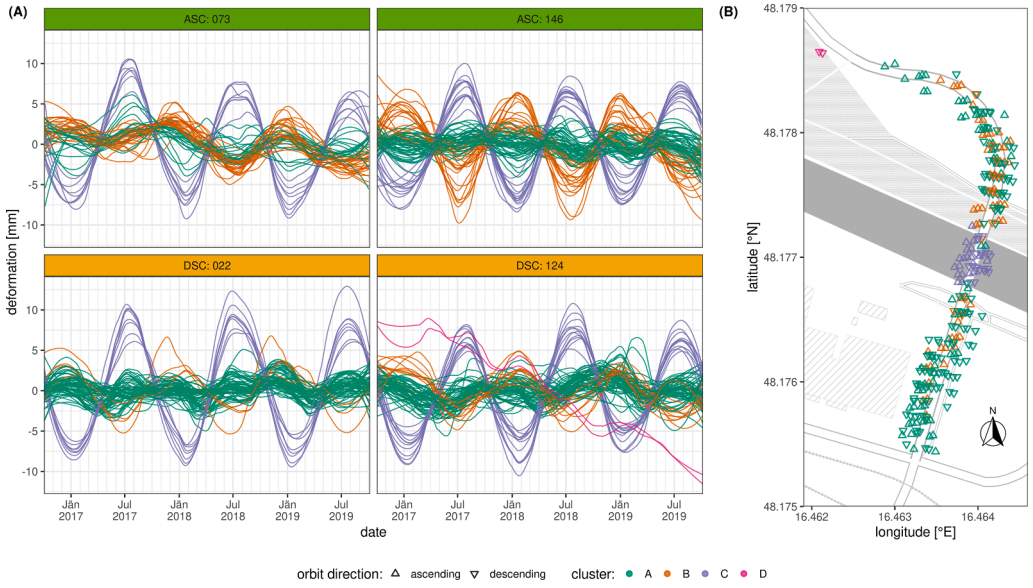


Fig. 4. Time series clustering. (A) depicts time series of all four orbits as well as their assignment to one of the four clusters. (B) shows the location of the temporal clusters in space.

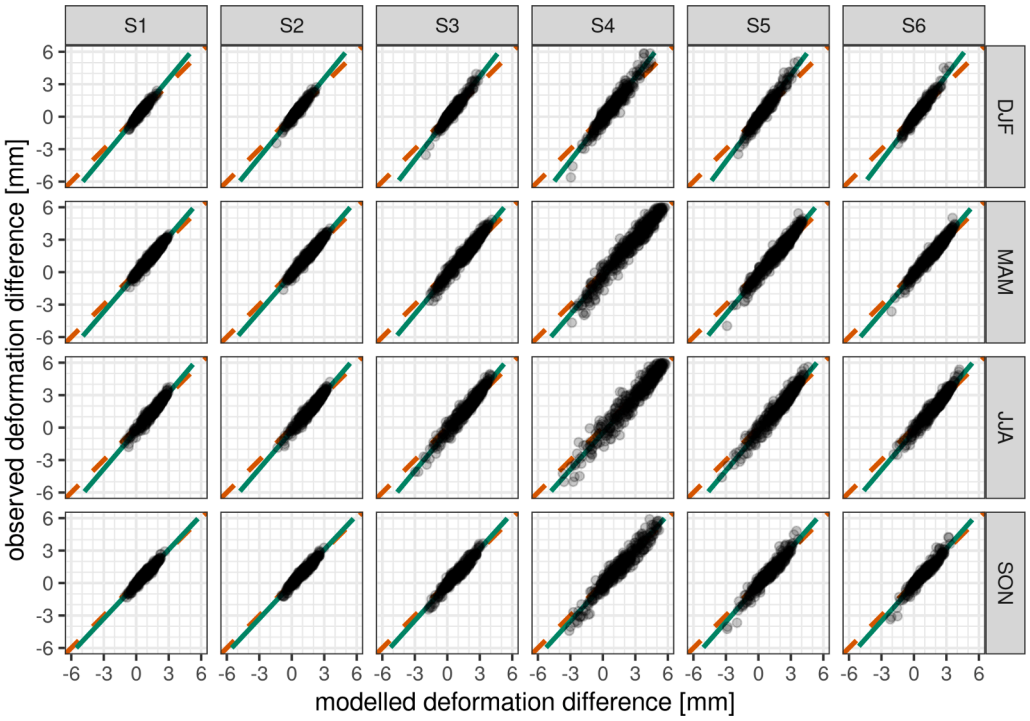


Fig. 5. Comparison of predicted versus observed deformation values. The orange line depicts x = y, i.e. a 1:1 fit. The green line, which shows a linear regression of observed values on predicted values, indicates a slight bias of the model for large deformation differences between the two sensing times. Monthly values are grouped by season, with DJF (December, January February) representing winter, MAM (March, April, May) representing spring, JJA (June, July, August) representing summer and SON (September, October, November) representing fall. A plot at monthly resolution is provided as supplement 5. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Notably, the bias was lowest during fall.

With respect to variable importance of the meteorological features, relative humidity (afternoon) and air temperature (both afternoon and at night) prevailed as most prominent variables (Fig. 6), followed by the zero degree level (both afternoon and night), global radiation (afternoon) and sunshine duration (afternoon). Global radiation at night and precipitation during both periods possessed little to none explanatory power.

5. **Line-of-sight to vertical**: Conversion from line-of-sight deformation to vertical and horizontal deformation components could be performed for 12,294 grid points on the bridge. For these points, time series of daily deformation rates for both horizontal and vertical (east/west) components were available from 2016–10-02 to 2019–10-06 on a daily basis (i.e. 1,100 days).
6. **Time series clustering II**: Both vertical and horizontal deformation component series were clustered using *k*-means for longitudinal data in order to identify consistent deformation patterns within the final combined deformation values. The approach was identical to (3) described above, only the input data set was different. The optimal number of clusters was determined to be $k = 3$ for vertical and $k = 4$ for horizontal deformation.

Clustering of vertical deformation rates yielded three distinct clusters (Fig. 7). The most pronounced pattern could be observed for points belonging to cluster C. This cluster exhibited distinct seasonal deformation, namely upwards (+z) in summer and downwards (−z) in winter. In terms of spatial distribution, the cluster consistently represented the main part of the bridge above the Danube Canal. Cluster B did also show a seasonal deformation pattern, albeit less pronounced and in opposing direction, i.e. downwards in summer and upwards in winter. A majority of points belonging to this cluster were located around the northern dilatation joints of the bridge, as well as some smaller areas along the northern and southern ramps. Finally, cluster A could be interpreted as noise without a distinct pattern. Points assigned to this cluster were found on both ramps, especially on the southern one.

In terms of horizontal deformation, four distinct clusters were identified (Fig. 8). Clusters A and B do again represent noise. Notably, cluster A showed slight indications of seasonal effects in late summer 2018 and to some extent also in 2019. Cluster B was mainly located on the northern and southern ramps of the bridge. Points belonging to cluster A were distributed across the whole area. Clusters C and D showed pronounced seasonal patterns: While cluster C exhibited eastwards (positive) deformations during summer, the smaller cluster D showed westwards (negative) deformations during summer. To some extent, this was also reflected with respect to their spatial location, with cluster C being located on the Eastern and cluster D on the Western part of the bridge.

7. **Aggregation of results**: Two main auxiliary conditions were considered for deriving the final ensembles of deformation patterns. First, cluster assignments for time series of both longitudinal and vertical deformation component series were considered. Second, only time series within a radius of 40 meters (2 pixels) around the in situ sensor locations were selected and assigned to the nearest sensor (Fig. 9).

By combining these two different pieces of information, clean and robust ensembles of deformation series could be derived for the main part of the bridge. The merit of using time series clusters in addition to spatial distance to in situ measurement points was highlighted in both vertical (Fig. 10) and horizontal (Fig. 11) components: While general bridge deformation patterns can be observed within the whole data set of available time series, results are distorted or even superimposed by noisy deformation series.

A combination of information on both time-series cluster and closest in situ sensor resulted in distinct patterns of both vertical (Fig. 12) and horizontal (Fig. 13) deformation over time. In particular, it could be observed that clusters C (vertical) and D (horizontal) are solely present on the central and southern parts of the bridge around sensors S3 to S6. Sensor S2 was represented by the smallest number of clusters with respect to both horizontal and vertical deformation components.

Finally, a comparison of the derived ensembles of vertical deformation component series with in situ measurements provided by the water level gauge measurements yielded promising results (Fig. 14). The figure shows the original in situ deformation measurements in violet, and smoothed in situ measurements in green. Ensembles of Sentinel-1 based InSAR-results are displayed in orange, with the bold

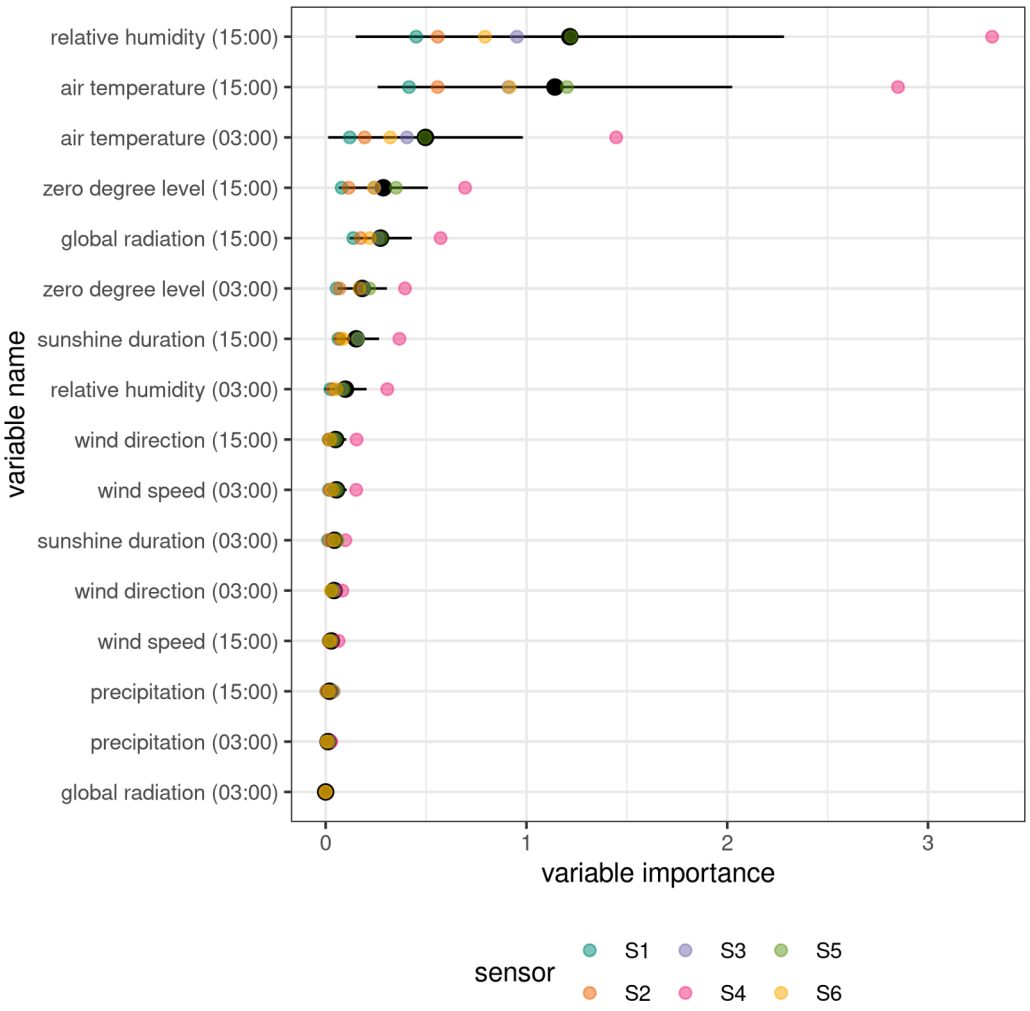


Fig. 6. Random forest variable importance (permutation importance) for all weather variables used. Black points and ranges indicate the overall mean and standard deviation across all sensors. Results for all single sensors are included as further reference, with colors corresponding to sensor name. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

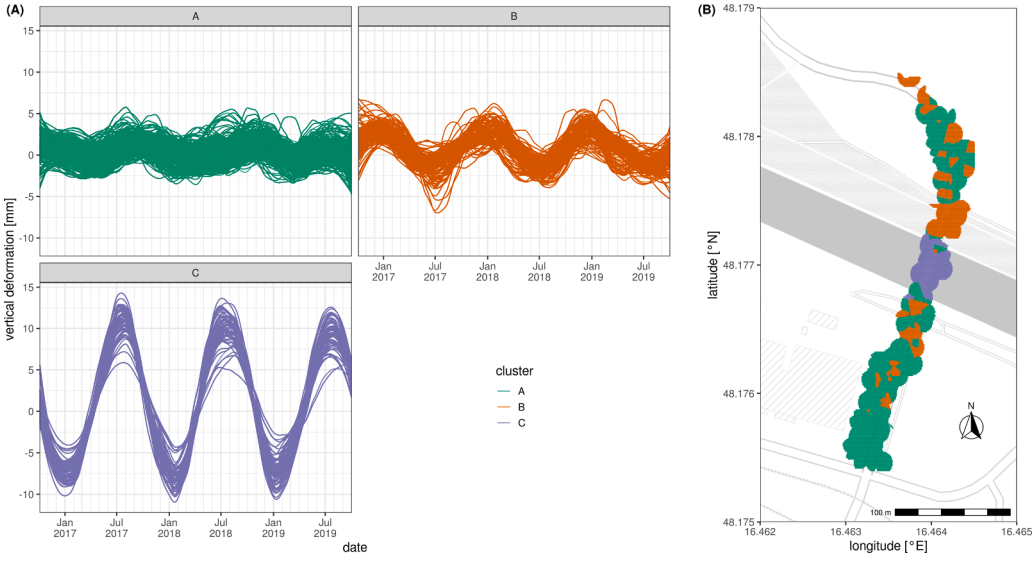


Fig. 7. Time series clusters for vertical deformation component (A) and corresponding location of cluster points on the bridge (B).

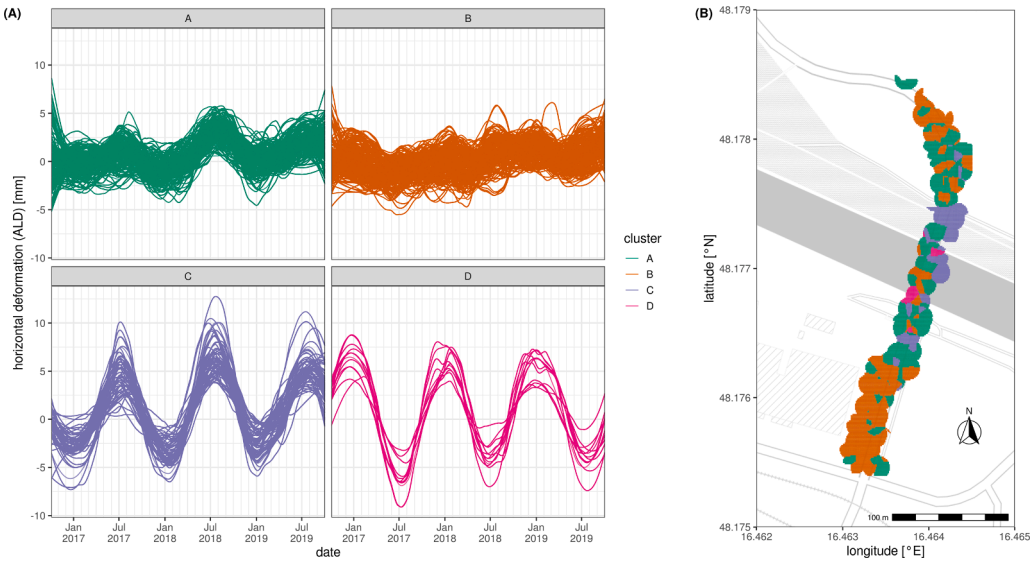


Fig. 8. Time series clusters for horizontal deformation component (A) and corresponding location of cluster points on the bridge (B).

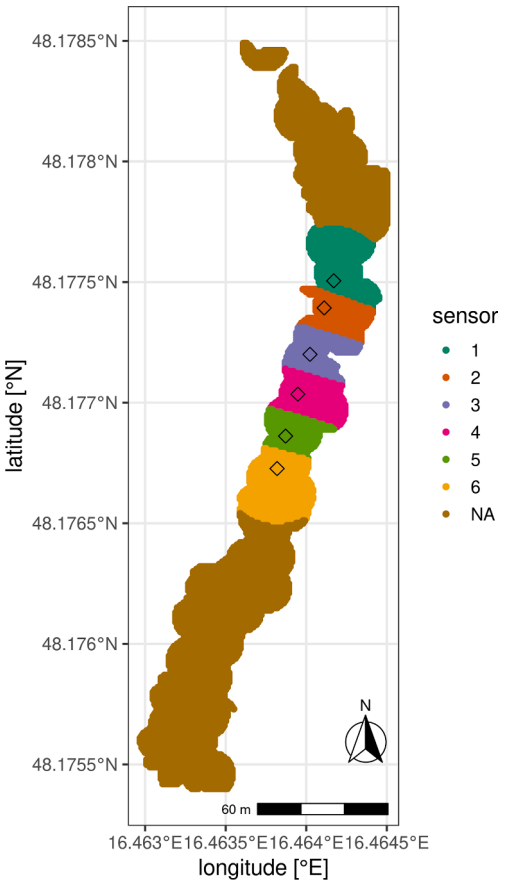


Fig. 9. Assignment of valid grid points on the bridge to their associated in situ sensor positions (i.e. the closest sensor within a radius of 40 m).

line indicating the ensemble mean, and the shaded orange ribbon indicating the minimum–maximum span of all ensemble members. PSI results for sensors S3 to S6 are represented through cluster C, S2 is represented through cluster B, and clusters A plus B were used for S1.

Results on the southern part of the bridge (sensors S5 and S6) were remarkably accurate, providing vertical deformation rates completely consistent with in situ measurements. Results around the bridge center were adequate as well. While S3 shows a consistent ensemble that slightly overestimates the deformation rates, results for S4 were biased in the opposite direction. Results derived via multitemporal InSAR captured the overall seasonality quite well, but underestimate the deformation rate by 5 to 10 mm. Results obtained for the northern part of the bridge were less satisfactory. In-situ measurments of S2 showed no particular pattern. InSAR results for this area were somewhat more sinuous than the observations, with a more pronounced seasonal pattern, which could not be detected in the in situ measurements. S1, which showed a weak but distinct seasonal pattern in the in situ measurements, is basically only represented as noise in the InSAR results.

8. **Time series decomposition**: STL decomposition illustrated the absolutely dominating seasonal effect (Fig. 15). Interpretation of the trend-component in a sub-mm range was not feasable and indicated the absence of a dedicated trend during the time period under investigation.

4. Discussion

Overall, results exceeded expectations in terms of quality and accuracy, especially when taking the comparably low spatial resolution (with respect to the target application) of Sentinel-1 SLC products into account. Our findings showed the general feasibility of using multitemporal InSAR analysis based on Sentinel-1 data for infrastructure monitoring. This is mainly attributed to the extensive post-processing workflow presented in this study. Benefits and drawbacks of the applied steps are discussed in the following. For the sake of clarity, certain single steps of the post-processing workflow are discussed jointly.

4.1. Time series cleaning and smoothing

Time-series cleaning and smoothing proved to be essential steps to tidy the original PSI time series derived via StaMPS. First, certain sensing dates produced exceptionally high deformation rates (spikes) across multiple scatterer points. If left uncorrected, these clear outliers could lead to unwanted effects and eventually misinterpretations further on in the analysis due to their leverage effect. The method based on using Friedman's Super Smoother for non-seasonal series and a robust STL decomposition for seasonal series works exceptionally well. While a quick manual double check of the results is advisable, results showed a

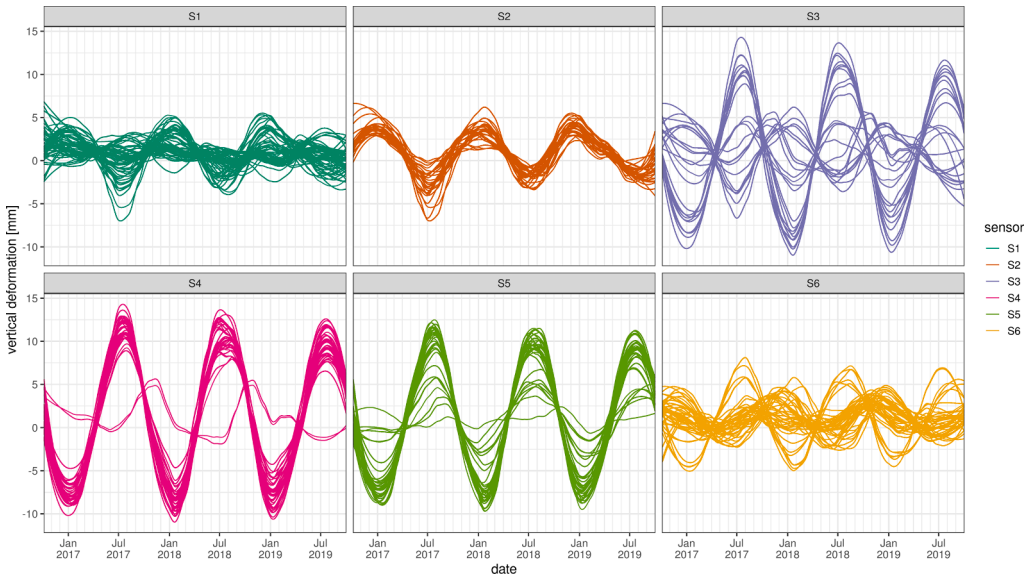


Fig. 10. Vertical deformation component time series grouped by closest in situ sensor (S1–S6).

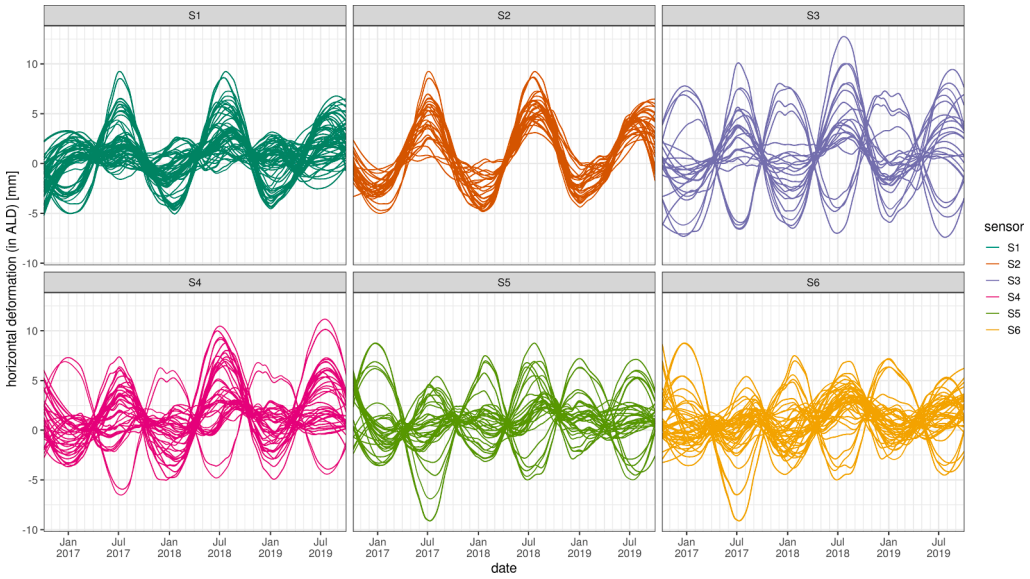


Fig. 11. Horizontal deformation component time series (in ALD) grouped by closest in situ sensor (S1–S6).

reasonably conservative outlier removal with only clear outliers being flagged. Its good performance on both seasonal and non-seasonal time series render the method universally applicable to any type of InSAR time series.

Second, the signal-to-noise ratio of the original StaMPS time series was suboptimal in many of the original PSI time series. The application of an appropriate filter thus helped to reduce the noise within each single time series. This cleaning step is recommended after outlier removal in order to avoid aforementioned distorting effects (bumps) of outliers in the smoothed series. The comparison of different smoothers revealed that the choice of the smoothing algorithm was not of prime importance, as long as the parameters of the smoother are controlled appropriately. The only exception was smoothing via GAMs. While this is computationally efficient, results were slightly worse (i.e. too much generalisation) than in all other cases.

Ultimately, the exact sources for both spikes and noise remained unclear. The most likely explanation seemed to be that these artefacts were attributable to atmospheric and weather effects, which remained despite the thorough choice of StaMPS parameter settings and the application of methods for atmospheric correction.

Apart from noise reduction, another important benefit of time series smoothing was the possibility to obtain time series on a daily basis instead of the original time interval of six days. This alleviated many further post-processing steps, and was particularly useful for a consistent derivation of vertical and horizontal components on a daily basis from multiple orbits with different acquisition dates.

4.2. *Bias correction for different sensing times*

The approach for bias correction was developed especially for the assessment of deformation rates on bridges. Even more specifically, the approach described here is tailored to this particular bridge structure and the circumstances regarding its monitoring. Albeit this step is probably not required for every type of critical infrastructure – since the effects stemming from different sensing times can simply be neglected for some structures – it is argued that this is particularly important for

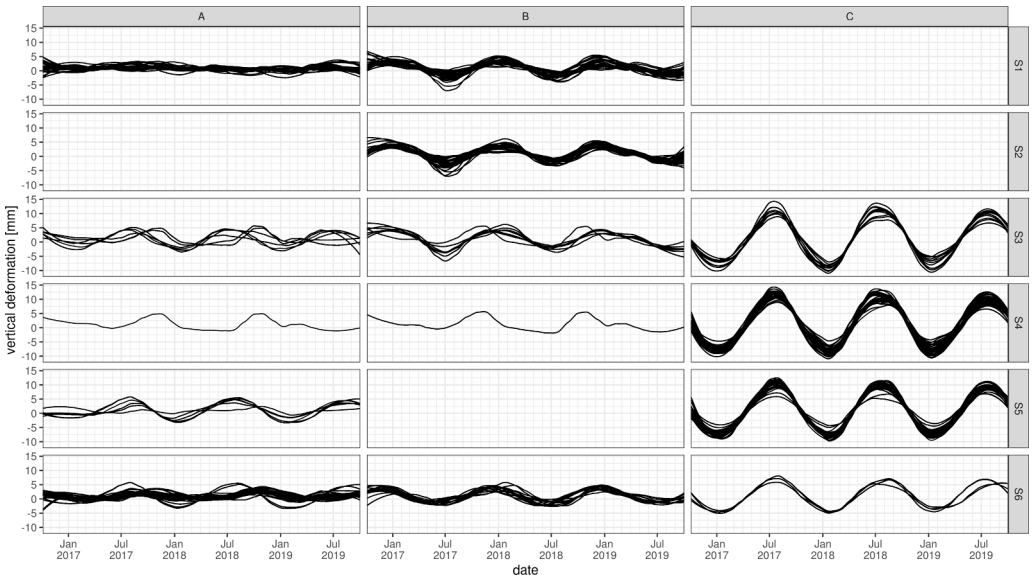


Fig. 12. Vertical deformation component grouped by time series cluster (A-C, columns) and in situ deformation sensor (S1–S6, rows).

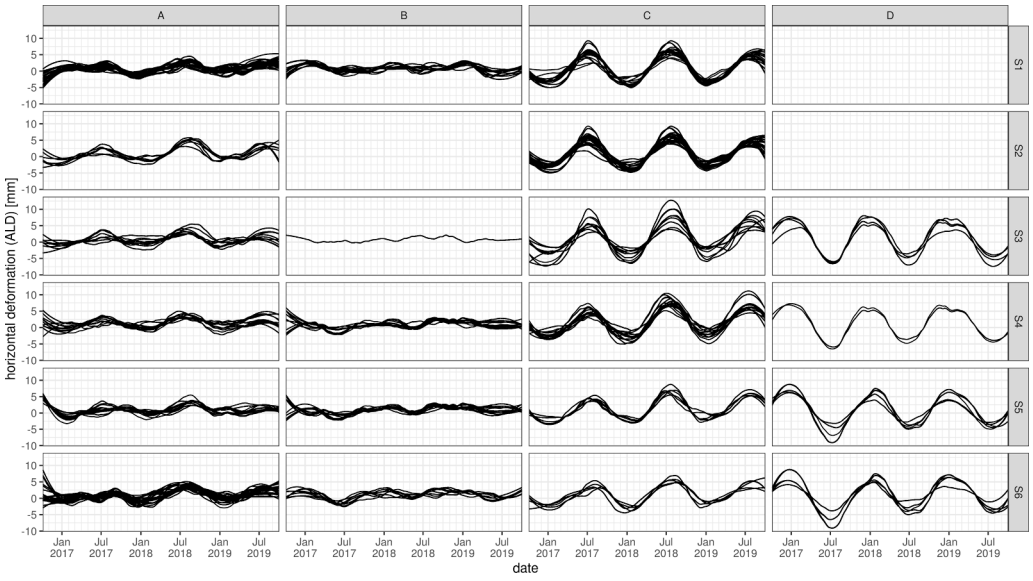


Fig. 13. Horizontal deformation component in ALD grouped by time series cluster (A-D, columns) and in situ deformation sensor (S1-S6, rows).

structures that undergo a distinct diurnal deformation cycle of substantial magnitude that changes over the course of a year.

A clear drawback of the multivariate modelling method applied here is the amount of data required. Apart from high resolution weather data, in situ deformation measurements over a certain period of time are needed to train a robust model. While weather data are needed for the whole period of observation, the temporal coverage of in situ-measurements (training data) should cover at least two to three years to ensure basic requirements with respect to robustness.

In addition, one might rightfully interpose that bridge monitoring via InSAR is obsolete if the bridge is monitored using in situ sensors anyways. However, bridges are usually designed to last decades, and maintenance of in situ sensors is resource-intensive. Given the total lifetime of such structures, a comparatively short measurement campaign over several years will suffice to train a sound model. At the example of *Seitenhafenbrücke*, maintenance of the water level gauge and preparation of measurement data had only been budgeted for a limited period of time. Yet, available data covering more than five years were sufficient to derive an accurate model with solid goodness-of-fit. Also, if no in situ measurements were available, data derived from bridge deformation simulations or at least empirical estimates from structural engineers derived from experience with other bridges might have been used instead.

It is therefore advocated that this effect is accounted for if (1) the problem is expected to occur at the structure under consideration, and (2) if deriving an appropriate model is possible within reasonable expense of resources. The exact design of the model is of course subject to the specific structure under consideration.

As for the multivariate modelling approach used here, random forests prevailed as convincing method for such purposes. They are comparatively easy to handle, computationally efficient, and requirements in terms of hyperparameter tuning and statistical assumptions about data properties are low. At the same time, they are capable of modelling potentially complex relationships and interactions between variables. Results showed that accuracy in (sub-) millimetric range could be achieved for deformation differences of up to 7 mm by using several

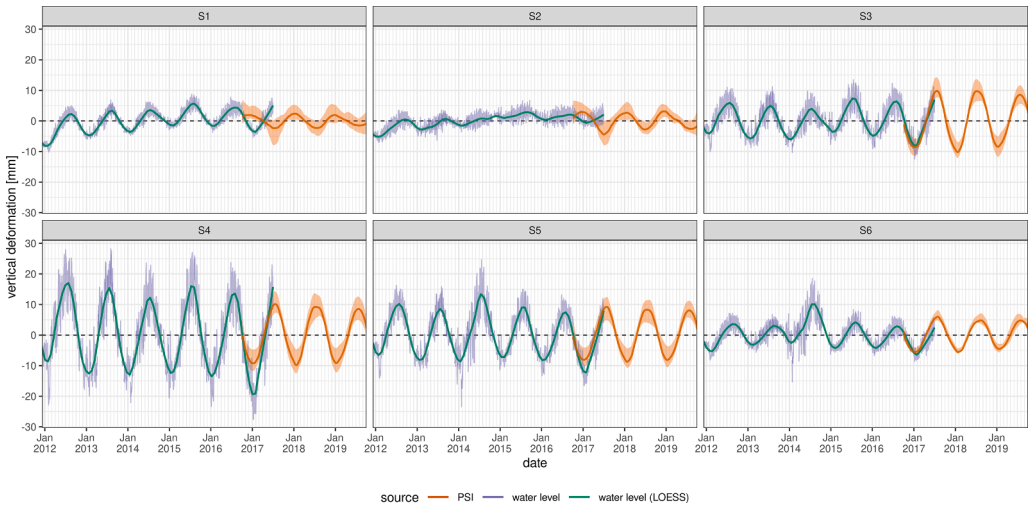


Fig. 14. Comparison of final vertical deformation component time series derived from Sentinel-1 SLC with measured in situ data. For comparative purposes, all series are plotted with mean zero.

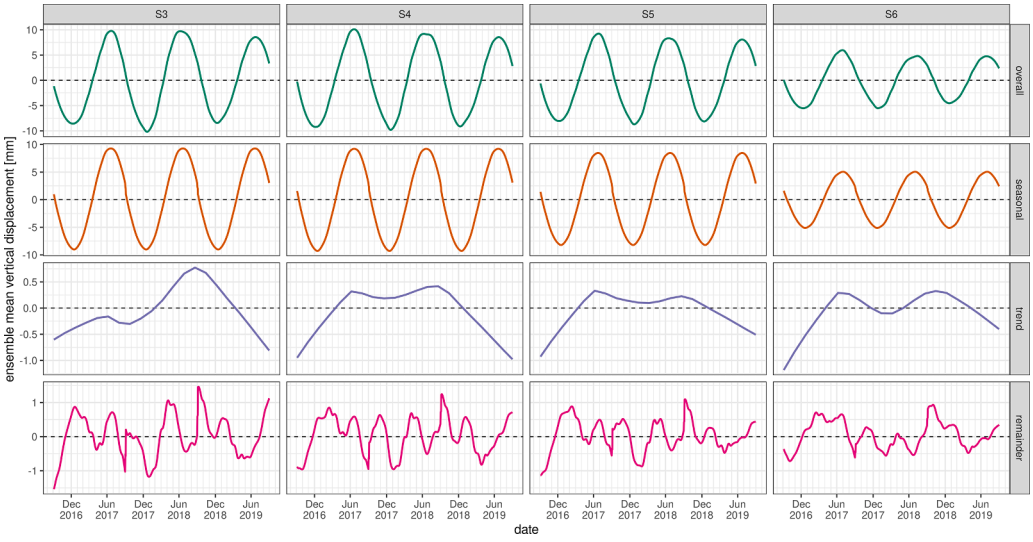


Fig. 15. Decomposition of the time series representing the ensemble mean into seasonal, trend and noise components.

weather variables as explanatory features. The slight bias towards underestimating singular, particularly high deformation differences is probably difficult to alleviate. Nevertheless, this is also an indicator that the underlying model is robust and not susceptible to overfitting.

With respect to variable importance, the relevance of air temperature and relative humidity among all meteorological parameters used does seem perfectly plausible.

4.3. Derivation of vertical and horizontal deformation components

The final aggregation procedure applied to obtain final results consisted of multiple, complementary considerations and processing steps: The creation of a formally defined grid covering the whole area of interest, the extraction of vertical and horizontal deformation components from line-of-sight vectors, the derivation of time series clusters and the consideration of data within close spatial proximity.

The creation of a fine raster across the area of interest (the whole bridge in this case) proved to be a good approach to achieve maximum coverage of the target object while circumventing the problem of where to locate the resulting component vectors for vertical and horizontal deformation. Since PS points are located at slightly different positions in each of the orbits due to observation geometry, identified PS points cannot be compared unequivocally with respect to their 'true' position. In the case of structures with a complex surface (i.e. with a multitude of potential persistent scatterers) such as *Seitenhafenbrücke*, a definite determination of the exact location of the scatterer points is not possible. By using a pre-defined grid, the extraction of vertical and horizontal components from line-of-sight data was easily automated by simply combining time series from the closest ascending and descending orbit, respectively. Given the coverage by two orbits in both orbit directions, this procedure also resulted in combinations of data from different orbits.

The use of time series clustering was an essential step to detect distinct patterns in order to separate interesting scatterer points from noisy ones. Albeit the selection of the optimal number of clusters turned out to be slightly cumbersome (as it required some in-depth scrutiny of results obtained for different numbers of clusters), longitudinal clustering clearly improved the interpretability of results. The demonstration that locations with comparable deformation patterns could be identified by clustering deformation time series without considering spatial information in the process was an interesting finding. This has implications way beyond infrastructure monitoring and may be

potentially useful for any kind of satellite-based InSAR analysis.

Eventually, the aggregation of raster points within a certain distance around in situ sensors allowed the generation of ensembles of deformation vectors. This turned out to be beneficial, as not only some mean deformation could be derived. Instead, the use of empirical quantiles of the underlying ensemble members allowed to establish an intrinsic measure of uncertainty for each time stamp. The choice of sensor locations as centers for spatial aggregates was taken primarily for practical purposes, since in situ sensors were used for validation. In order to assess the general deformation behaviour of a bridge, other selections of cluster centers might have been equally suitable. Ultimately, location of cluster centers and the definition of a radius for aggregation are strongly dependent on the use case.

As far as interpretability and validity of results is concerned, the comparison with in situ measurements indicated a viable performance of the whole analysis. The general deformation patterns were well detectable in the resulting InSAR-based deformation series. Results turned out to be quite concordant for three out of the six sensor locations (S3, S5 and S6), all of which are located on the middle part of the bridge (i.e. the river structure across the Danube Canal). Deformations occurring around sensor S4, which is also located on the middle structure, were captured adequately. However, the magnitude of the deformation rates was underestimated at the middle of the bridge. To some extent, this might be attributable to the applied smoothing procedure of the time series, in combination with the signal noise. It is conceivable that the smoothing led to a slight underestimation of the peak deformations during summer and winter. A smaller span width might have alleviated this to some extent. However, this seemed inadvisable in the presented case due to the noise in the data. It is argued that improving atmospheric correction methods would lead to better results in this case. Congruence of InSAR-based in situ measurements was worse at the northern river structure around the locations of the remaining two sensors, S1 and S2. Albeit the magnitude of the deformations obtained from multitemporal InSAR processing was within expectations, the underlying deformation patterns differ slightly. While in situ measurements of S1 indicate a small but clearly detectable seasonal deformation, this is not discernable in our results. While cluster A indicated no deformation at all, cluster B even bore slight indications of an anti-cyclical deformation, with downward motion in summer and upward motion in winter. This is even more pronounced at the location of sensor S2. Reasons for this deviation remained unclear. However, it is crucial to note that the sensors are located on the bottom of the bridge, while PSI measurements are taken on the surface of the bridge. Therefore, these inconsistencies occurring on the northern river segment of the bridge might be attributable to the different locations where both measurement types are taken.

Interpretation of horizontal deformation components was more difficult, since no reference measurements were available. While the four time series clusters are quite consistent, the location of the points belonging to each cluster was not as distinct as in the case of vertical deformation components. Due to the favorable location of the bridge, deformations projected on the horizontal plane in azimuth look direction correspond almost to a 'left–right' deformation when considering a cross section of the bridge. To some extent, it was assumed that clusters C and D represented thermal expansion of the bridge during summer, which was not only reflected by an upward/downward deformation in the vertical axis, but does also coincide with slight simultaneous deformations towards both east and west (expansion) during summer, and inwards deformation (contraction) during winter.

Altogether, results do seem plausible, and validation with in situ measurements indicated solid performance, at least regarding vertical deformation. Further analysis and discussion with structural engineers might provide deeper insight into this topic.

4.4. Time series decomposition

While STL decomposition does not unearth any surprising effects at the example of *Seitenhafenbrücke* in the port of Vienna, the dominating superimposition of effective deformation trends is of utmost interest for discussion. Deformation rates from multitemporal InSAR analyses are usually reported as mean deformation over a certain period of time, assuming a consistent linear trend of the underlying deformation process. While this might arguably be feasible for certain applications such as slope creep, assuming a linear trend in complex time series is an oversimplification that might lead to erroneous conclusions.

In the present case, starting and end point were deliberately chosen so that the period under consideration comprised exactly three years. In addition, boundaries of the series are located in fall, where bridge deformation is close to its mean. If no attention is paid to the selection of start and end point of the time period under consideration, seasonal maxima and minima towards either end of the series might lead to unwanted artefacts. With respect to using simple linear regression to obtain average deformation rates over time, extreme values towards the endpoints of the time series scatter plot might significantly alter the slope of the fitted regression line due to their leverage effect. At the same time, poor choice of start and end dates does also affect the workflow described here. For instance, applying general filter techniques to series exhibiting seasonal extreme values at the beginning or end might result unreliable filter estimates at the end points.

It is therefore advocated that trends in seasonal time series should be properly derived by applying some sort of seasonal decomposition of the time series.

4.5. Transferability to other domains

Our findings showed the general feasibility of using multitemporal InSAR analysis based on Sentinel-1 data for infrastructure monitoring. By means of extensive post-processing, initial results derived PSI analysis via SNAP and StaMPS can be refined significantly. Eventually, applying stratification by both temporal and spatial clusters to clean time series yields consistent and clear results.

The applicability of the proposed workflow – or at least of parts of it – is valid well beyond the example presented herein. In terms of sensors, the procedure clearly generalises beyond Sentinel-1. Similar post-processing steps can be applied to any sensor providing data for satellite-based, multi-temporal SAR interferometry. In terms of application, proposed working steps are not only limited to critical infrastructure monitoring, but are also transferable to other domains. Processing steps like time series cleaning and smoothing, bias correction for different sensing times, time series clustering or STL-decomposition are universally usable for many analyses involving remotely sensed time series.

5. Conclusion

Deformation monitoring of critical infrastructure assets is of major importance for preventing damage and losses. The availability of C-Band SAR data from ESA's Sentinel-1 satellites open up new potential for infrastructure monitoring using InSAR techniques. Despite some restrictions in terms of spatial resolution and a sub-optimal signal-to-noise ratio, Sentinel-1 based multitemporal InSAR analyses can lead to promising results if appropriate post-processing is applied.

Suggested post-processing steps include time series cleaning and smoothing to reduce outliers and noise. A bias correction to account for small but systematically occurring deformation between different sensing times is of particular importance when working with infrastructure assets that potentially undergo some diurnal deformation cycle that changes over time, such as certain bridges which might exhibit seasonal cycles in daily deformation patterns. Extracting horizontal and vertical deformation components from satellite line-of-sight data by using multiple orbits is of core importance to practitioners and structural engineers for further assessment of the results. Aggregation of results by combining longitudinal clustering and spatial grouping of points in close

vicinity allows to obtain robust ensembles of deformation patterns. Despite these promising results, more research is required to test the transferability of the proposed approach to other types of critical infrastructures. Yet, this research provides essential blueprints that may be used for further exploration of the potential of (not only Sentinel-1 based) multitemporal InSAR analysis for critical infrastructure monitoring.

Author contributions

MS initiated this work and conceived the presented idea. The PSI analysis was conducted by BW and MS. Post-processing (time-series analysis, statistical programming) and visualisation was designed and performed by MS. MS wrote the manuscript. MA did proof-reading.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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In support of endeavours to avoid contentious terms the authors would like to point out that ‘leader’ and ‘supporter’ were used as alternative for the hitherto commonly used master/slave terminology throughout this manuscript.

Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at https://doi.org/10.1016/j.isprsjprs.2020.12.001.

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